Discrete Elastic Rod Optimization of Torsional Hinge Stiffness for Solar Array Deployment

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I. INTRODUCTION

Deployable solar panels are commonplace in most satellites today. For example, the Starlink constellation of satellites uses deployable solar arrays to allow for satellites to be installed into a launch vehicle in a compact configuration without compromising the area available for electricity generation once placed in orbit. These structures are typically folded into compact configurations and deployed in orbit using stored elastic strain energy, often relying on compliant mechanisms like tape-spring hinges.

Simulating this deployment is computationally challenging due to large nonlinearities and general mathematical complexity of modeling transient, high-speed systems. Thus, this project leverages the relatively simple Discrete Elastic Rods (DER) framework to model a 1D cross-section of an unfolding multi-panel solar array. The goal is to write a simulation that optimizes the torsional stiffness and required damping of the deployment hinges. The main constraints are the structural integrity of the supposed satellite and the force required to guarantee full deployment; if the hinges are too stiff, the array snaps open with excessive force and damages the spacecraft bus, but if the hinges are too pliant, the array fails to fully deploy.

II. RELATED WORK

The basic idea behind the Discrete Elastic Rods method was introduced in a paper by Bergou et al. (2008), which discretizes smooth curves to model large bending and twisting behaviors.

Specifically for deployable space structures, Pedivellano and Pellegrino (2021) investigated the deployment dynamics of flexible and lightweight thin-shell structures. They argue for the need to model folds as non-linear elastic hinges and also provided both computational and physical models to support the conclusions presented in the research. Burgoon et al. (2006) simulated the creation of origami using discrete methods to model the inelastic deformation and transient behavior of the folds, a precedent applicable to multiple engineering applications of plastically deformed folded joints. To study the locking mechanisms of deployable arrays on satellites, Saad, Desoki, and Kassab (2026) used implicit time-stepping to handle heavy computation and ensure numerical stability during large rotational deformations. Finally, Chen et al. (2019) utilized numerical optimization on a rotational symmetric model containing flexible joints to maximize the efficiency of the expansion process, using a similar framework to the one proposed in this project for a different geometry.

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III. PHYSICAL MODEL

To simplify the geometry as much as possible, only a cross section of the array will be modeled. The model will use the discrete elastic rod approach due to the cross-section only containing one dimension. Each long, rigid rod will represent a solar panel, while the hinges will be modeled as shorter, more flexible rods with a lower modulus of elasticity. The mass of each link will be approximated at the start of the simulation with the hinges being significantly lighter than the panel segments. The system will be represented by N nodes with N-1 edges constraining it. The twist angle at each of the nodes will be calculated. Additionally, the total elastic and bending energy will be calculated and minimized. The leftmost node will be set as the zero and fixed.

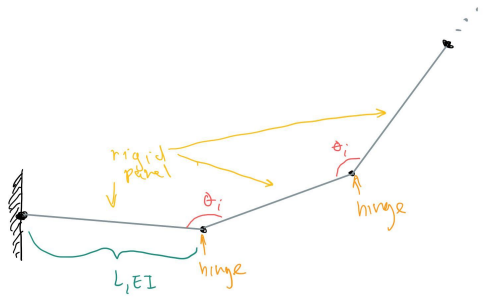


Fig. 1. Rough diagram of physical setup with three rigid segments and two hinge segments.

IV. ALGORITHM

The simulation will utilize an implicit Euler scheme and Newton-Raphson solver to handle time-stepping. A loop will be written to handle optimizing the bending stiffness (EI_{hinge}) and inherent damping of the hinge, with the goal of minimizing structural damage done to the satellite bus or hinge during deployment and upon locking. The loop will be written to achieve a critically damped target lockout velocity, v_{target}.

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The nodal forces are derived from the gradients of the system's elastic energy E_{total}, while damping is approximated as viscous drag with a coefficient c:

$$\mathbf{F}_{\text{elastic}} = -\nabla E_{\text{total}}(\mathbf{x}) \tag{1}$$

$$\mathbf{F}_{\text{damp}} = -c\mathbf{v} \tag{2}$$

In order to actually implement the results of the simulation into a physical design, a damper would likely need to be designed to meet the desired damping coefficient. For the purposes of this project, it will simply be set as a constant with no regard as to the physical feasibility of that damping value, though a literature search will be done in order to find an approximate range in which the value should lie.

- **Initialization**
  - Define physical parameters: positions, masses, stiffness, and damping
  - Set the initial configuration by assigning curvatures
  - Set the target state by assigning curvature
  - Define target: a final velocity
- **Optimization Loop**
  - Initialize a guess for the hinge bending stiffness EI_{hinge}.
  - Begin the implicit time-stepping loop
    - Calculate elastic and damping forces
    - Formulate the implicit Euler step
    - Solve for nodal positions using the Hessian of the total energy
    - Update nodal positions for the current time step.
  - Evaluate the final state and record the actual end velocity
  - Compute the error $e = v_{\text{final}} - v_{\text{target}}$ and update the stiffness EI_{hinge} guess

V. EXPECTED RESULTS AND VALIDATION

The simulation will output the stiffness of the optimized hinge, as well as a time history of positions and kinetic and potential energy over time within the hinge(s). Ideally, potential energy decreases over time, and kinetic energy starts and ends at zero.

To validate the code, the results will compared against existing literature mentioned in the Introduction section.

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